



判断题

1-1

While inserting n elements into a skew heap, the insertion of the last element may cost $O(n)$ time.

1-2

Insert 1, 3, 4, 9, 2, 5, 7, 8, and 6 one by one into an initially empty AVL tree. Then the preorder traversal of the tree must be {3, 1, 2, 5, 4, 7, 6, 8, 9}.

1-3

When solving a problem with input size N by divide and conquer, if at each stage the problem is divided into 9 sub-problems of equal size $N/3$, and the conquer step takes $O(N^2 \log N)$ to form the solution from the sub-solutions, then the overall time $O(N^2 \log^2 N)$.

1-4

In the Turnpike Reconstruction problem, when trying to place the new point at $x=d$, we only need to remove from the remaining distance multiset the distances between this point and the two endpoints (i.e., d and $L-d$). If both distances can be removed, we may continue; otherwise, this branch fails.

1-5

In a red-black tree, the length of the longest path from the root to a NIL leaf is at most twice the length of the shortest such path.

1-6

Dynamic programming computes each subproblem every time it is needed, without reusing any previously stored results.

1-7

In a B+ tree, internal nodes store both index keys and actual data records.

1-8

A binomial queue of size 842 can be represented as the binomial trees B_1, B_3, B_5, B_8, B_9

1-9

In inverted file index, we should index every words, even if they are so common that almost every document contains them(i.e "a", "the", "it").

1-10

While deleting node with value v on a splay tree, the new root value may be the largest value which is smaller than v .

选择题

2-1

What is the time complexity of an algorithm whose running time $T(n)$ is described by the recurrence $T(n) = 9T(n/3) + O(n^2)$?

- A. $O(n^2)$
- B. $O(n^2 \log n)$
- C. $O(n^3)$
- D. $O(n^2 \log^2 n)$

2-2

Insert {7,6,5,4,3,2,1} into an initially empty splay tree, then find 5. Which one of the following statements is FALSE?

- A. 5 is the grandparent of 4
- B. 2 and 6 are siblings
- C. 3 is the right child of 2
- D. 1 is the left child of 2

2-3

If there are 15 nodes in an AVL tree, and there's at most one node with balance factor not equal to 0, then the maximum depth of the tree is _____. The depth of an empty tree is defined to be 0.

- [A. 4](#)
- B. 5
- C. 6
- D. 7

2-4

There are 40000 documents in the database. The statistic data for one query are shown in the following table. The precision is: ____

	Relevant	Irrelevant
Retrieved	3000	12000
Not Retrieved	7000	18000

- A. 30%
- [B. 20%](#)
- C. 40%
- D. 33%

2-5

To build a skew heap, we can start with an empty heap, and [merge](#) each single-node heap into the resulting heap one by one.

Then the best description of the time complexity of this procedure is:

- **A. $O(N \log N)$**
- B. $O(\sqrt{N})$
- C. $O(\log N)$
- [D. \$O\(N\)\$](#)

2-6

In a binomial queue with 180 nodes, how many nodes have [depth 1](#) (the root has depth 0)?

- [A. 18](#)
- B. 4
- C. 38

- D.19

2-7

Consider a dynamic array (i.e., a sequence list) with an initial capacity of $c=1$ and an initial size of $s=0$. It supports two operations:

push(x): Insert an element at the end. If $s = c$, the array is resized to $2c$, and all existing elements are moved to the new array.

pop(): Remove the last element (guaranteed that $s > 0$ when called). If after deletion $s = c/2$, the capacity is set to $c/2$, and all elements are moved to the new array.

The basic cost of a normal insertion or deletion is **1**.

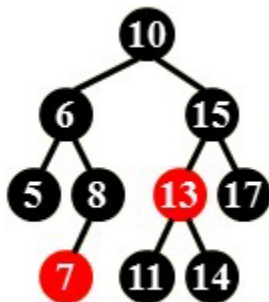
Each resizing operation (expansion or contraction) incurs a cost equal to **the number of elements moved** (i.e., proportional to the current size).

Question: For any sequence of N operations consisting of legal push and pop calls, what is the tight asymptotic bound of the amortized worst-case cost per operation under this implementation?

- A. $O(1)$
- B. $O(\log n)$
- C. $O(\sqrt{n})$
- D. $O(n)$

2-8

After deleting 10 from the red-black tree given in the figure, which one of the following statements must be FALSE?



- A. 8 is the parent of 6, and 6 is black
- B. 6 is the parent of 8, and 8 is red

- C. 11 is the parent of 15, and 15 is black
- D. 13 is the parent of 14, and 13 is black

2-9

Consider ordered sequences composed of prime numbers (repetition allowed) such that the sum of the elements equals exactly 12. Sequences differing in order are considered distinct; for example, (2, 3, 7) and (7, 2, 3) are counted as two different sequences. What is the total number of such sequences?

Hint: Consider building up the answer incrementally (starting from sums of 1, 2, 3, ..., up to 12) and use previously computed results to help determine the count for larger sums.

- A. 32
- B. 35
- C. 36
- D. 40

2-10

Insert 10, 20, 30, 40, 50, 60, 70, 80 into an initially empty 3-order B+ tree (i.e., each internal node can hold at most 2 keys). After all insertions, which of the following statements is TRUE?

- A. There are 3 leaf nodes
- B. The root node contains key 60
- C. The height of the tree is 3
- D. The leftmost leaf node contains keys [10, 20, 30]

3 程序填空题

Description

Bob has two children, and he wants to divide n pieces of food between them. The i -th piece of food has a nutrition value of a^{**i} .

Bob wants to be as fair as possible, so he hopes that the difference between the total nutrition values received by the two children is as small as possible.

Please help Bob find the minimum possible difference.

Input

The first line contains an integer T — the number of test cases.

Each test case starts with an integer n ($1 \leq n \leq 20$) — the number of food items.

The next line contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 1000$), where a_i is the nutrition value of the i -th food.

Output

For each test case, output a single integer — the minimum possible difference between the total nutrition values of the two children.

Input Sample

```
1
4
1 6 11 5
```

Output Sample

```
1
```

```

#include <stdio.h>
#include <math.h>

const int inf = 1000000000;
const int maxn = 20;

int n, total, sum, ans;
int a[maxn];

void solve(int p) {
    if (p == n) {
        int cur = _____//这里填空
        if (cur < ans) ans = cur;
        return;
    }
    sum += a[p];
    solve(p + 1);
    _____// 这里填空
    solve(p + 1);
}

int main() {
    int T;
    scanf("%d", &T);

    while (T--) {
        scanf("%d", &n);
        total = sum = 0;
        for (int i = 0; i < n; ++i) {
            scanf("%d", &a[i]);
            total += a[i];
        }
        ans = inf;
        solve(0);
        printf("%d\n", ans);
    }

    return 0;
}

```